

AegisAI Project Plan: Human Behaviour as a Security Boundary, Affective AI and Behavioural Analytics for Cybersecurity

- Operationalisation of the Human-Boundary Ecosystem (HBE)
 - Adaptive Edge AI Cybersecurity Awareness Training Platform for SMEs
 - Version: 1.0
 - Status: Formal Implementation Roadmap — Phase 2 (System Operationalisation)
 - 4 weeks Sprints Iterations
 - Date: 25 February 2026
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Introduction

The AegisAI vision depends on an integrated and technologically core ecosystem of four foundational pillars, each powered by AI and Machine Learning. These pillars, the AI Engine, the Human Firewall Initiative, Automated Adversarial Simulation, and a commitment to Trust by Design, create a self-reinforcing, adaptive security system.

This document formalises the transition from research to operational deployment of the human-centric, regulation aware an edge deployed adaptive cybersecurity components, constructing the base ground for this part of the AegisAI platform. This document will describe the bootstrapping of the core components that will build the Human-Boundary Ecosystem, mainly the components that focus on tracking user behaviour, understanding users emotion driven behavior using the software components and the development of the Adaptive Edge AI Cybersecurity Awareness Training Platform.

The solution is centralised around four core AI-driven experiments for the AI component, with a privacy-by-design and on-edge architecture, to ensure raw personal behavioural data never leaves the client side. Mainly to be aligned with the main regulations like the EU AI Act, NIS2 Directive, and GDPR, while delivering a solution for the employee cyber resilience through continuous, personalised microlearning.

The implementation is structured into seven Work Packages (WP1—WP7) that cover data strategy, platform architecture, edge processing, regulatory compliance, and enterprise-grade data governance.

In Summary the core components that will be addressed in this implementation plan are the following: human-centric approaches in software design and interactions, this involve the

design of software components in terms of UI/UX and AI components to track and understand human interactions; regulation-compliance in terms of how we build and design the overall system and how we handle client data and client systems; an edge-native adaptive cybersecurity training ecosystem that turns mandatory awareness training into a proactive, privacy-preserving risk-management asset for SMEs.

The following table summarises the four core experiments we can address at the end of this implementation plan. It covers both ML models that can be used and statistical experiments that we can run to collect and analyse data.

Experiment	Objective	Inputs	AI Technique	Outputs	SME Use Case
Behavioural Drift Detection	Detect changes in employee cyber behaviour, indicating increased vulnerability	Interaction logs, response latency, click paths, error patterns	Time-series anomaly detection, concept drift detection	Drift score, risk adaptation flag	Identify at-risk employees and trigger timely microtraining
Gamified Cyber Awareness Profiling	Build individual cyberbehaviour profiles from training interactions	Game decisions, scenario choices, risk assessment answers	Behavioural clustering, risk scoring	Cyber behaviour profile	Generate personalised training paths
Emotional / Feedback Tracking	Monitor engagement and emotional state to prevent training fatigue	Feedback text, self-reported stress/confidence, interaction speed	Sentiment analysis, affective inference	Engagement index, stress/overload indicator	Adapt learning pace and prevent burnout
Adaptive Content Optimization	Dynamically select optimal training scenarios	Performance history, drift signals, emotional index	Reinforcement learning / multi-armed bandit models	Personalised scenario selection	Deliver continuous, maximally effective microlearning

In the following sections, we segment the development of this project into workpackages.

WP1 — AI Experiments & Behavioral Modeling Framework

Overview

Design, formalize, and validate four AI-driven experiments that model human cybersecurity behavior, emotional engagement, adaptive training effectiveness, and behavioral drift in SME environments.

This WP defines the scientific backbone of the platform.

Scope

Formal experiment definition, model selection, evaluation metrics, SME alignment.

- Define inputs, outputs, and modelling techniques
- Formalize evaluation metrics
- Align models with SME operational needs
- Ensure explainability and regulatory alignment

Tasks

Task ID	Task Name	Description	Output
T1.1	Experiment Formalization	Define objectives, inputs, AI techniques, outputs, and SME use-case mapping for four experiments.	AI Experiment Specification Document
T1.2	Model Selection & Evaluation	Select ML techniques (anomaly detection, clustering, sentiment analysis, reinforcement learning), define evaluation metrics and explainability mechanisms.	Model Evaluation Framework
T1.3	SME Applicability Modeling	Translate model outputs into actionable dashboards, risk indicators, and compliance evidence.	SME AI Integration Report

Deliverables

Deliverable	Description
D1.1	AI Experiment Specification
D1.2	Model Evaluation & Validation Framework
D1.3	AI Risk & Explainability Draft

WP2 — Data Modeling & Human Boundary Ecosystem Schema

Overview

Design a structured data model that supports behavioral monitoring, emotional modeling, adaptive learning, and compliance tracking.

Provides a structured foundation for AI, UX, governance, and compliance.

Scope

Data taxonomy, schema definition, privacy-aware structuring.

- Define structured schema
- Categorize data sensitivity
- Map UX events to data structure

Tasks

Task ID	Task Name	Description	Output
T2.1	Data Taxonomy Definition	Define behavioral, contextual, affective, and system metadata categories.	Data Taxonomy Document
T2.2	Schema Design	Define data structures, formats, metadata fields, retention tags. • Define JSON/relational structures • Define field-level documentation • Include retention metadata	Data Schema Specification
T2.3	Privacy Structuring	Classify personal vs pseudonymized data; define minimization and anonymization rules, align with GDPR and related regulation.	Privacy Classification Matrix

Deliverables

Deliverable	Description
D2.1	Data Schema Specification
D2.2	Data Lifecycle Map
D2.3	Data Sensitivity, minimization and anonymisation Report

WP3 — Gamification Platform UI/UX Design & User Experience Validation

Overview

Design a gamified cybersecurity training system serving as a behavioural data engine. Primary mechanism for behavioural data acquisition and adaptive learning. Ensures engagement, ethical data capture, and compliance and transparency.

Scope

Scenario design, risk scoring logic, and admin monitoring structure, with wireframes, prototypes, interaction modelling, and usability validation.

Tasks

Task ID	Task Name	Description	Output
T3.1	Scenario Design	Develop phishing simulations, micro-challenges, and risk evaluation tasks.	Scenario Blueprint
T3.2	Behavioral Event Mapping	Define UX events mapped to the data capture structure.	Event Capture Map
T3.3	Admin Monitoring Design	Define risk dashboards, engagement analytics, department-level insights.	Admin Monitoring Model
T3.4	Persona & Journey Mapping	Define employee/admin personas and usage journeys.	Persona Documentation
T3.5	Interaction Modeling	Define state transitions, event triggers, behavioral capture points.	Interaction Flow Diagrams
T3.6	Wireframe Design	Develop low-fidelity designs for training and admin flows.	Wireframe Set
T3.7	Interactive Prototyping	Build clickable prototype simulating adaptive logic.	Clickable UX Prototype
T3.8	Usability Testing	Conduct engagement, cognitive load, and trust validation.	UX Validation Report

Deliverables

Deliverable	Description
D3.1	Gamification Design Blueprint
D3.2	Behavioural Data Capture Model
D3.3	Training Effectiveness Metrics

D3.4	Persona & Journey Maps
D3.5	Wireframes & Prototype
D3.6	UX Validation Report
D3.7	UX-to-Data Mapping Document

WP4 — Edge Architecture & Privacy-by-Design Infrastructure

Overview

Field	Description
Objective	Design secure edge-based architecture ensuring local processing and privacy preservation.
Strategic Role	Core privacy differentiation and regulatory compliance enabler.
Scope	Edge deployment, encryption, secure data flow, federated learning (optional).

Tasks

Task ID	Task Name	Description	Output
T4.1	Edge Node Architecture	Define local microservices, database design, encryption mechanisms.	Edge Architecture Blueprint
T4.2	Secure Data Flow	Design internal pipelines and anonymization layers.	Data Flow Diagram
T4.3	Federated Learning Strategy	Define optional global model improvement without raw data sharing.	Federated Learning Framework

Deliverables

Deliverable	Description
D4.1	Edge Deployment Architecture
D4.2	Security & Encryption Specification
D4.3	Privacy-by-Design Technical Documentation

WP5 — Regulatory Compliance & Governance Alignment

Overview

Field	Description
Objective	Align system with GDPR, EU AI Act, and NIS2 Directive.
Strategic Role	Legal robustness and market credibility.
Scope	Risk classification, logging, explainability, compliance reporting.

Tasks

Task ID	Task Name	Description	Output
T5.1	AI Risk Classification	Determine risk level and mitigation measures.	AI Risk Assessment
T5.2	Transparency Framework	Design user and admin explainability layers.	Transparency Model
T5.3	Compliance Documentation	Define logging, audit mechanisms, and regulatory documentation.	Compliance Mapping Matrix

Deliverables

Deliverable	Description
D5.1	Regulatory Mapping Matrix
D5.2	AI Risk Management Framework
D5.3	Audit & Evidence Package Template

WP6 — SME Operational Integration

Overview

Field	Description
Objective	Define real-world SME integration model.
Strategic Role	Ensures practical usability and adoption.
Scope	Onboarding workflows, periodic training cycles, compliance reporting.

Tasks

Task ID	Task Name	Description	Output
T6.1	Deployment Modeling	Define onboarding, configuration, and deployment scenarios.	Deployment Guide
T6.2	Operational Workflow	Model periodic training cycles and reporting routines.	Workflow Diagrams
T6.3	Adoption Strategy	Define SME integration roadmap and training model.	Adoption Framework

Deliverables

Deliverable	Description
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D6.1	SME Deployment Guide
D6.2	Operational Workflow Documentation
D6.3	Compliance Reporting Model

WP7 — Data Management & Governance Framework

Overview

Field	Description
Objective	Establish structured data governance for platform and client environments.
Strategic Role	Ensures sustainability, compliance, and operational clarity.
Scope	Data lifecycle, metadata standards, quality monitoring, retention, rights management.

Tasks

Task ID	Task Name	Description	Output
T7.1	Data Inventory & Classification	Catalogue datasets and classify sensitivity levels.	Data Catalogue
T7.2	Metadata & Documentation	Define data dictionaries, lineage tracking.	Metadata Repository
T7.3	Data Quality Monitoring	Implement validation rules and quality metrics.	Data Quality Dashboard
T7.4	Retention & Deletion Policies	Define retention timelines and automated deletion rules.	Retention Policy
T7.5	Data Subject Rights Management	Define processes for access, correction, deletion, portability.	Rights Management Workflow

Deliverables

Deliverable	Description
D7.1	Data Governance Policy
D7.2	Data Catalogue & Metadata Repository
D7.3	Data Quality & Retention Documentation
D7.4	Data Subject Rights Framework

Final Workpackage Structure

WP	Domain
WP1	AI Modeling
WP2	Data Schema
WP3	Gamification System UI/UX
WP4	Edge Architecture
WP5	Regulatory Compliance
WP6	SME Integration
WP7	Data Governance